

1. (1) Let $(X_1, X_2, \dots, X_n)$ be a random sample of size n taken from a Normal population with parameters: mean θ_1 , and variance θ_2 . find the maximum likelihood estimates of these two parameters

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

(Sol. Maximum likelihood Estimate (MLE) for normal population

Let random sample $X_1, X_2, \dots, X_n$ from a normal dist $N(\mu, \sigma^2)$ where $\mu = \theta_1$ and $\sigma^2 = \theta_2$

(1) likelihood function

the likelihood function $L(\theta_1, \theta_2)$ is the product of the individual prob density functions:

$$\begin{aligned} L(\theta_1, \theta_2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\theta_2}} \exp\left(-\frac{(x_i - \theta_1)^2}{2\theta_2}\right) \\ &= (2\pi\theta_2)^{-n/2} \exp\left(-\sum_{i=1}^n \frac{(x_i - \theta_1)^2}{2\theta_2}\right) \end{aligned}$$

(2) log likelihood function

taking natural logarithm ($\ln$):-

$$l = \ln L = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\theta_2) - \frac{1}{2\theta_2} \sum_{i=1}^n (x_i - \theta_1)^2$$

(3) Estimating θ_1 (mean)

Take partial derivative wrt θ_1 and set to 0.

$$\frac{\partial \ell}{\partial \theta_1} = -\frac{1}{2\theta_1} \sum_{i=1}^n 2(x_i - \theta_1)(-1) = -\frac{1}{\theta_1} \sum_{i=1}^n (x_i - \theta_1) = 0$$

$$\sum_{i=1}^n x_i - n\theta_1 = 0 \Rightarrow \hat{\theta}_1 = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

(4) Estimating θ_2 (variance)

Take the partial derivative wrt θ_2 and set to 0

$$\frac{\partial \ell}{\partial \theta_2} = -\frac{n}{2\theta_2} + \frac{1}{2\theta_2^2} \sum_{i=1}^n (x_i - \theta_1)^2 = 0$$

Multiply by $2\theta_2^2$:

$$-n\theta_2 + \sum_{i=1}^n (x_i - \hat{\theta}_1)^2 = 0 \Rightarrow \hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Q2 let $X_1, X_2, \dots, X_n$ be a random sample from $B(m, \theta)$ distribution, where $\theta \in \Theta = (0, 1)$ is unknown and m is a known +ve integer. Compute value of θ using MMSE Estimation

Sol. MMSE (Min mean square error)

We have sample $X_i \sim \text{Binomial}(m, \theta)$ where m is known

Let $Y = \sum_{i=1}^n X_i$. Since the sum of independent Binomial variables with same θ is also binomial, $Y \sim \text{Binomial}(nm, \theta)$

→ the MMSE estimator of a parameter θ given data Y is the posterior mean: $E[\theta | Y]$. To compute this,

we need a prior distribution θ . Assuming a uniform prior $\theta \sim \text{Beta}(1, 1)$ [or $U(0, 1)$], which is common when no prior information is provided.

① Posterior Distribution

The likelihood is $f(y|\theta) \propto \theta^y (1-\theta)^{nm-y}$

The posterior is $\pi(\theta|y) \propto f(y|\theta) \cdot \pi(\theta) = \theta^y (1-\theta)^{nm-y} \cdot 1$

This is the kernel of Beta $(y+1, nm-y+1)$ distribution.

② MMSE Estimate

The mean of a Beta (α, β) distribution is $\frac{\alpha}{\alpha+\beta}$.

$$\hat{\theta}_{\text{MMSE}} = E[\theta|y] = \frac{y+1}{(y+1) + (nm-y+1)} = \frac{y+1}{nm+2}$$

substitute $y = \sum x_i$:

$$\hat{\theta}_{\text{MMSE}} = \frac{\sum_{i=1}^n x_i + 1}{nm+2}$$